

Muhammad Rehan

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Education

- 2023– **Graduate Student**, Program in Indo-European Studies, UCLA
- 2023–2024 **M.A. in Indo-European Studies**, UCLA
- 2022–2023 **Post-Baccalaureate Certificate in Classics** (with Distinction), UCLA
- 2018–2022 **B.A. in Classics (Hons.) and German Studies (Hons.)**, Grinnell College

External Coursework and Certifications

- 2026 Advanced Bayesian Data Analysis, Ninth Summer School on Statistical Methods for Linguistics and Psychology, Potsdam
- 2025 Linguistic Typology in Evolutionary Perspective, Statistical Measures in Corpus Linguistics, Foundations of Predictive Modeling with R, Variationist Sociolinguistics, Reconstructing Syntax, Adaptive Speech Perception and Comprehension, Language Phylogenies, LSA Summer Institute, University of Oregon
- 2024 Exceptionality, Phonological Grammar and Opacity, Historical Linguistics, CreteLing 2024
- 2021 Old English, Leiden Summer School in Languages and Linguistics
- 2021 The Ancient Greek Workshop, UC Berkeley
- 2020 Deutsch Erleben Workshop, Humboldt Universität
- 2019 German Language Program, Universität Leipzig

Research Interests

Quantitative historical linguistics; probabilistic models of sound change; linguistic phylogenetics; Bayesian spatiotemporal modeling; Agent-based modeling; Indo-European philology and linguistics.

Publications

Peer-Reviewed Articles and Articles under Review

- Under Review* **Muhammad Rehan**, Valentina Lunardi, and David Goldstein, “**Supervised measurement of Christian Latin.**”
Digital Scholarship in the Humanities
- 2026 Grant S. McCall and **Muhammad Rehan**, “**E-lorgesailie: An agent-based simulation model of Acheulean lithic raw material procurement and handaxe deposition patterns.**”
Journal of Archaeological Method and Theory 33.49.

- 2021 Grant S. McCall, Russell Greaves, Brian Hitchcock, Brian Ostahowski, Sherman W. Horn III, and **Muhammad Rehan**, “**The estuarine ecological knowledge network: Future prospects.**”
Marine Technology and Science Journal 55 (3).122–123.

Conference Proceedings

- Forth.* **Muhammad Rehan**, “Re-evaluating the word-medial operation of the Saussure-Hirt Effect in Greek.”
Proceedings of the 4th International Conference on Ancient Greek Linguistics.
- 2026 **Muhammad Rehan**, “Gk. **μοιχός** ‘adulterer’: Re-mixing a piss-poor etymology”
Proceedings of the 35th Annual UCLA Indo-European Conference Vol. 35.147–163.

Presentations & Posters

Quantitative and Computational Linguistics

- 2026 **Muhammad Rehan**, “Positional asymmetries in consonantal sound change.”
5th AMC Symposium: “Types of Evidence in Historical Linguistics”, Dec. 9.
- 2026 **Muhammad Rehan** and David Goldstein, “Language contact is not consistently associated with increased feature economy.”
Typology in Historical Phonology Workshop 2026, Aug. 25.
- 2026 Tom Francis and **Muhammad Rehan**, “Modeling alphabet diffusion and community interactions in ancient Italy with spatiotemporal point processes.”
Connected Past 2026, Sept. 23.
- 2026 **Muhammad Rehan** and David Goldstein, “Spatiohylogenetic modeling provides new insights into feature economy.”
qc-leste: Quantitative Cross-Linguistic Explorations of Space, Time, and Evolution, Jun. 10.
- 2026 Tom Francis and **Muhammad Rehan**, “Point processes and spatiotemporal modeling of alphabet spread in Ancient Italy.”
DataX, UCLA, May 19.
- 2026 **Muhammad Rehan**, “Why do languages have the sounds they do: Insights from infant experiments and phylogenetic modeling.”
DataX, UCLA, April 20.
- 2026 **Muhammad Rehan**, “Discerning segmental stability using phylogenetic models.”
Southern California Annual Meeting on Phonology, Apr. 11.
- 2026 **Muhammad Rehan** and David Goldstein, “From points to polygons: Modeling spatial autocorrelation.”
Special workshop on “The geography of linguistic evolution”, EVOLANG XVI, Apr. 7.
- 2025 **Muhammad Rehan**, Sean Liu, and Marc Canby, “Converging computational evidence against an East-Germanic clade.”
WeCIEC 36: Annual UCLA Indo-European Conference, Oct. 28.

- 2025 **Muhammad Rehan, “Population size has a weak effect on phoneme inventory size.”**
Poster presented at the *Linguistic Society of America Summer Institute, University of Oregon*, Jul. 10.
- 2024 **Muhammad Rehan, “Towards an integrative approach to the investigation of linguistic phylogenies.”**
5th Munich-UCLA Historical Linguistics Colloquium, Jun. 28.
- 2024 **Muhammad Rehan, “FSAs, FSTs and sequence alignment in phylogenetic studies: The case of Tocharian B.”**
UCLA Program in Indo-European Studies Graduate Seminar, May 29.

Phonetics

- 2027 **Muhammad Rehan and Coralie Cram, “Vowel-to-vowel coarticulation is language specific and modulated by phonology.”**
LSA Annual Meeting 2027, Jan. 8.
- 2027 **Muhammad Rehan and Megha Sundara, “Infants’ discrimination of sound pairs predicts their occurrence in the world’s languages.”**
Five-Minute Linguist, LSA Annual Meeting 2027, Jan. 8.
- 2026 **Muhammad Rehan, and Megha Sundara, “A tool to derive P-map constraints from infants’ discriminability of phonetic contrasts.”**
Annual Meeting on Phonology, Oct. 9.
- 2026 **Muhammad Rehan, Jahnavi Narkar, and Dhanya Charan, “Lexical stress and syllable weight in Hindi-Urdu pop text-setting.”**
Linguistic Society of America Annual Meeting 2026, Jan. 9.
- 2025 **Muhammad Rehan, Jahnavi Narkar, and Dhanya Charan, “Investigating lexical stress in Hindi-Urdu pop music.”**
Sixth Joint Meeting of Acoustical Society of America and Acoustical Society of Japan, Dec. 5.

Phonological Change and Indo-European Philology

- 2026 **Muhammad Rehan, “The phonetic value of Avestan <ā>.”**
American Society for Premodern Asia Annual Meeting, Apr. 24.
- 2025 **Muhammad Rehan, “The development of word-final stops in Hittite.”**
AnatoLinks, University of Oxford, Dec. 4.
- 2025 **Muhammad Rehan, “Synchronic phonetic variation drives diachronic sound change: Further evidence from pre-nasal raising in Young Avestan.”**
7th Edinburgh Symposium on Historical Phonology, Dec. 2.
- 2025 **Tom Francis, Klaus Kim, Laura McLean, and Muhammad Rehan, “Issues of lexical cognacy encoding in IE-CoR.”**
27th International Conference on Historical Linguistics, Aug. 18.
- 2025 **Muhammad Rehan, “Re-evaluating the word-medial operation of the Saussure-Hirt Effect in Greek.”**
11th International Colloquium on Ancient Greek Linguistics, June 26.

- 2025 **Muhammad Rehan, “Word-final voicing in Hittite?”**
Southern California Annual Meeting on Phonology 2025, Apr. 12.
- 2024 **Muhammad Rehan, “Re-mixing a piss-poor etymology: On Gk. *μοιχός* and OE *(ge)māh*.”**
WeCIEC 35: Annual UCLA Indo-European Conference, Oct. 25.

Digital Philology

- 2026 Elisa Migliaretti, Spiridon-Iosif Capotos, **Muhammad Rehan**, and Zachary Fletcher,
“MoPSoS: Introduction to the database of Ancient Greek morphology, prosody, syntax and scansion.”
“Epic through Numbers” panel, Celtic Conference in Classics, Jul. 15.

Guest Lectures

- 2026 **“Why are some words more stable than others? Bayesian modeling of lexical replacement in Indo-European.”**
CLS-185: Origins and Nature of English Vocabulary, UCLA, Jul. 7.
- 2026 **“How can experimental psycholinguistics and diachronic phonology help us learn languages?”**
LING 114: How to Learn a Foreign Language, University of Arizona, March 14
- 2025 **“brms tutorial.”**
LING 217: Experimental Phonology, UCLA, Nov. 18
- 2025 **“Origins and nature of food vocabulary in English.”**
Classics 185: Origins and Nature of English Vocabulary, UCLA, Jul. 8.
- 2025 **“Historical linguistics.”**
LING-20: Introduction to the Study of Language, UCLA (with Aidan Holmgren and Nick Guymon).
- 2024 **“Karma is the breeze in my hair on the weekend: Sanskrit loanwords into English.”**
Classics 185: Origins and Nature of English Vocabulary, UCLA, Jul. 11.

Teaching

- Summer 2026 **Instructor of Record**
 LING-20: Introduction to Linguistic Analysis, UCLA
 Faculty Supervisor: Prof. Stefan Keine
- Spring 2026 **Teaching Assistant**
 LING-110: Introduction to Historical Linguistics, UCLA
 Instructor of Record: Dr. Thomas Motter
- 2026 **Teaching Assistant**
 CLS 52: Greco-Roman Technologies: Materials, Monuments, Machines
 Instructor of Record: Prof. Sarah Beckmann

- 2025 **Teaching Assistant**
CLS 51: Art and Archaeology of Ancient Greece
Instructor of Record: Prof. John Papadopoulos
- 2025 **Instructor of Record**
LING-110: Introduction to Historical Linguistics, UCLA
Faculty Supervisor: Prof. Stefan Keine
- 2025 **Undergraduate Supervisor**
LING-197: Infants' Discrimination of Phonetic Contrasts, UCLA
Faculty Supervisor: Prof. Megha Sundara
- 2025 **Teaching Assistant**
LING-20: Introduction to Linguistic Analysis, UCLA
Instructor of Record: Prof. Ben Eischens, Prof. Anoop Mahajan
- Fall 2024 **Teaching Assistant**
LING-110: Introduction to Historical Linguistics, UCLA
Instructor of Record: Prof. David Goldstein
- Sum. 2024 **Teaching Assistant**
LING-1: Introduction to the Study of Language, UCLA
Instructor of Record: Corrina Fuller
- Fall 2021 **Instructor**
ALS-102: Urdu, Grinnell College

Research and Academic Work Experience

- 2026– **Humanities Technology Consultant**
HumTech, UCLA
- 2026– **Research Associate**
Center for Human-Environmental Research
- 2025 **Research Assistant**
Supervisor: Prof. Brent Vine, UCLA
- 2023– **Research Assistant**
Supervisor: Prof. David Goldstein, UCLA
- 2020 **Research Assistant**
Supervisor: Prof. Grant McCall, Tulane University

Scholarships & Awards

- 2026 Keck Summer Graduate Fellowship
- 2025 Graduate Summer Research Mentorship Award
- 2024 & 2025 Morvarid Guiv Graduate Fellowship in Zoroastrian Studies

2024	UCLA/Keck Humanistic Inquiry Research Award (shared with Tom Francis, Klaus Kim, and Laura McLean)
2024	Graduate Summer Research Mentorship Award
2023–24	Institute for the Study of Global Antiquity Fellowship
2023	Society for Classical Studies Lionel Pearson Fellowship
2022–23	Joan Palevsky Chair Post-Baccalaureate Scholarship in Classics
2022	CAMWS Award for Outstanding Accomplishment in Classical Studies
2022	Gus and Elfie Kalfass Prize for Excellence in German Studies
2021	Gerald W. LaLonde Fellowship in Classical Studies
2021	UC Berkeley Intensive Greek Workshop Fellowship
2019	Institute for Global Engagement Summer Language Study Scholarship

Languages

Ancient	Advanced: Ancient Greek, Hittite, Gothic, Latin, Vedic Intermediate: Middle High German, Old Persian, Avestan, Luwian
Modern	(Near-)Native: Hindi/Urdu, Panjabi, English; German (C1)

Technical Skills & Programming Languages

Microsoft Office Suite, NetLogo, RevBayes, Beast2.0, Python, $\LaTeX$, Julia, Stan, R
R packages: `sp`, `sf`, `brms`, `lme4`, `mgcv`, `rstan`, `R-INLA`, `geostan`, `inlabru`, `cmdstanR` and many others

Service

2026–	Webmaster, Program in Indo-European Studies, UCLA
2026–	Seminar Organizer, DataX, UCLA
2025–	Abstract Reviewer, Southern California Undergraduate Linguistics Conference
2024–26	Graduate Seminar Organizer (with Aidan Holmgren), Program in Indo-European Studies, UCLA
2023–26	Mentor, Asian and Asian American Classical Caucus
2023–	Organizer, UCLA Indo-European Conference
2023–24	Graduate Student Association Representative, UCLA Undergraduate Council
2021–22	Co-Chair, Grinnell College Classical Studies Student Education Policy Committee
2020–22	Co-Chair, Grinnell College German Studies Student Education Policy Committee

Professional Organizations

- 2025– American Society for Premodern Asia
- 2025– Linguistic Society of America
- 2025– Acoustical Society of America
- 2023– Indogermanische Gesellschaft
- 2022– Society for Classical Studies
- 2022– Classical Association of Midwest and South

Miscellaneous Publications

- 2023 **Muhammad Rehan, “Mind your language, Mister! A case study in Aristophanic humor.”**
Animus Vol. 3.1 (Winter).36–51.
- 2023 **Muhammad Rehan, “Between need, want, material and innate beauty: Tracing shades of Erōs.”**
Corvus Vol. 13 (Spring).27–35.
- 2023 **Muhammad Rehan, “Gendered language in Aristophanes’ *Birds*.”**
Persephone Vol. 8.1 (Spring).12–29.

References

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